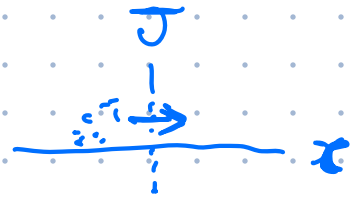
DIFFUSION IN 1d:

$c(x,t)$ - CONCENTRATION — PARTICLES PER UNIT LENGTH



$$\frac{\partial c}{\partial t} = -\frac{\partial}{\partial x} J$$

J - FLUX — PARTICLES PER UNIT TIME



$$J = -D\frac{\partial c}{\partial x}$$

$$\Rightarrow \boxed{\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}}$$

IN 2d AND 3d

$$\frac{\partial c}{\partial t} = -\nabla \cdot \vec{J}$$

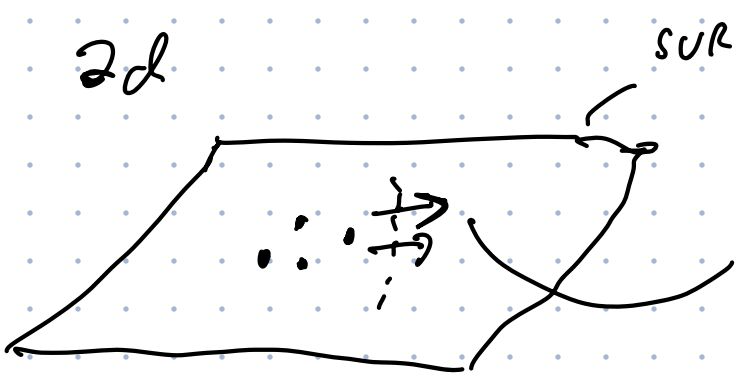
$$\vec{J} = -D\nabla c$$

$$\Rightarrow \frac{\partial c}{\partial t} = D\nabla^2 c$$

LAPLACIAN

IF D IS SPATIALLY UNIFORM

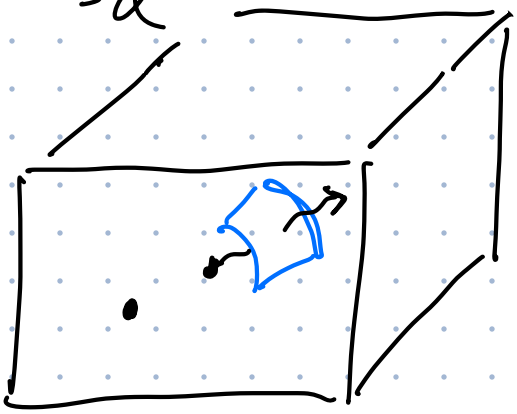
IN 2d



$\vec{J}$ - PARTICLES PER LENGTH PER TIME

c - PARTICLES PER AREA

IN 3d



$\vec{J}$ - PARTICLES PER AREA PER TIME

c - PARTICLES PER VOLUME

IN 1d, D HAS UNITS OF LENGTH^2/TIME. WHAT IS IT IN 2d OR 3d?

PS3

$D - \frac{\text{LENGTH}^2}{\text{TIME}}$ $\qquad \frac{\text{AREA}^2}{\text{TIME}} = \frac{\text{LENGTH}^4}{\text{TIME}}$

$\frac{\text{LENGTH}^3}{\text{TIME}}$

IN 2d

$$\frac{\partial c}{\partial t} = D\nabla^2 c$$

$$= D\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 c}{\partial \theta^2}\right)$$

$\frac{1}{\text{TIME}}$ $\qquad$ LENGTH^2

D HAS UNITS $\frac{\text{LENGTH}^2}{\text{TIME}}$

$\frac{\partial c}{\partial r} = ?$ $\qquad c(r,\theta) = R(r)\Theta(\theta)$

$$\nabla^2 c = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 c}{\partial \theta^2}$$ (0)

RADIAL SYMMETRY